

Instruction Interpretation & Sequencing (Self-Learning)

Q. Explain the concepts of Instruction Interpretation and Sequencing. How is the control flow of instructions maintained during execution?

> **Definition to Remember**

1. Instruction Interpretation

The process where the Control Unit determines the meaning of a fetched machine instruction.

2. Instruction Sequencing

The process of determining the memory address of the *next* instruction to fetch.

3. Modifying Control Flow

While the default flow is sequential, programs require loops and decisions. **Control Flow Instructions** alter the sequencing process by changing the PC:

4. Fetch-Decode-Execute Cycle

The synthesis of both concepts:

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

> `Inte
Branch